

Learned Z-Score Forecasting for Cross-Exchange Cryptocurrency Spreads: A Gradient-Boosting Approach on Live Microstructure Data

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ABSTRACT

Cross-exchange price spreads for identical cryptocurrency assets are widely documented, yet the literature universally relies on rule-based z-score thresholds to generate trading signals. We demonstrate that a gradient-boosting model (LightGBM) trained on live centralized-exchange microstructure features can predict the next-snapshot z-score of same-asset cross-exchange spreads, replacing mechanical threshold rules with a learned forecast. Machine learning enables the model to capture nonlinear interactions between orderbook imbalance, recent spread trajectory, and cross-venue dispersion that fixed threshold rules cannot exploit. The model ingests 68 features spanning orderbook imbalance, trade flow, funding rates, open interest, and lagged spread dynamics across 23 volatile assets and 6 exchanges at approximately one-minute granularity. These microstructure features require live collection and cannot be reconstructed from historical exchange APIs, making the resulting dataset a standalone contribution. A confidence filter trades only when the predicted z-score magnitude exceeds 0.5, selecting high-quality entries from the forecast distribution. In an 8-hour live paper-trading campaign, the filtered policy achieves directional accuracy of 76.9%, prediction R^2 of 0.35, and an hourly portfolio Sharpe ratio of 2.41, with positive risk-adjusted returns in every observed hour. Under matched capacity constraints, the learned policy improves per-trade directional accuracy by 7.6 percentage points and mean profit by 71% over a mechanical persistence baseline on identical settlement rules. We publicly release the cross-exchange microstructure dataset to enable community benchmarking. These results establish that supervised z-score forecasting with microstructure features offers a strictly stronger trading signal than the z-score threshold rules that underpin existing high-frequency cryptocurrency pairs-trading systems.

CCS CONCEPTS

• **Computing methodologies** → **Supervised learning by regression**; • **Applied computing** → *Economics*; • **Information systems** → *Data mining*.

KEYWORDS

statistical arbitrage, cryptocurrency, cross-exchange spreads, gradient boosting, market microstructure, limit order book, live evaluation

ACM Reference Format:

Anonymous Author(s). 2026. Learned Z-Score Forecasting for Cross-Exchange Cryptocurrency Spreads: A Gradient-Boosting Approach on Live Microstructure Data. In *Proceedings of 7th ACM International Conference on AI in Finance (ICAIF '26)*. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 INTRODUCTION

Cryptocurrency markets are fragmented across dozens of centralized exchanges (CEXs), each operating independent order books with heterogeneous liquidity, fee structures, and matching-engine latencies. Unlike equity markets, where consolidated tape systems and best-execution regulation enforce price convergence, crypto exchanges are isolated venues. This fragmentation creates persistent cross-exchange price deviations for identical assets [8].

Statistical arbitrage (“pairs trading”) exploits mean-reverting spreads between co-moving instruments [6]. In cryptocurrency markets, every existing pairs-trading study uses *rule-based z-score thresholds* to generate entry and exit signals: trade when the spread’s rolling z-score exceeds a fixed boundary, exit when it reverts [1, 5, 10, 14]. No prior work trains a model to *predict where the z-score will be next* and uses that forecast as the trading signal.

We demonstrate that a gradient-boosting model (LightGBM) trained on live centralized-exchange microstructure features can predict the next-snapshot z-score of same-asset cross-exchange spreads, replacing mechanical threshold rules with a learned forecast. The machine learning approach is motivated by the observation that cross-exchange spread dynamics depend on nonlinear interactions between orderbook imbalance, lagged z-scores, and cross-venue dispersion—relationships that fixed threshold rules are structurally unable to capture [11–13]. The model ingests 68 features spanning orderbook imbalance, trade flow, funding rates, open interest, and lagged spread dynamics across 23 volatile cryptocurrency assets and 6 exchanges at approximately one-minute granularity. A confidence filter restricts trading to predictions where $|\hat{z}| \geq 0.5$, selecting high-quality entries from the forecast distribution.

1.1 Dataset and Non-Backfillable Features

A distinguishing feature of this work is the dataset itself. Our live collector snapshots 9 concurrent signal families every 60–90 seconds across 23 assets and up to 6 exchanges: ticker quotes (best bid/ask), L2 orderbook (top 20 price levels per side), recent trades, funding rates, open interest, withdrawal status, exchange operational status, precomputed spread matrices, and OHLCV candles. The asset universe was selected for high cross-exchange spread volatility and sufficient liquidity across multiple venues. Collection spans over two months of live windows (June–July 2026) at one-minute granularity.

Public exchange APIs serve historical OHLCV candles, but *not* historical order books, quote-level snapshots, or trade prints. Every orderbook and trade-flow observation in our dataset exists only because it was collected live and cannot be reconstructed retroactively for any past window. This non-backfillable property means that the microstructure features central to our model are unavailable to researchers working with standard historical data sources.

The resulting dataset is released publicly through the public dataset release (URL withheld for double-blind review). To the best of our knowledge, no comparable public dataset provides synchronized cross-exchange L2 orderbook, trade flow, and funding rate data at minute-scale granularity for volatile cryptocurrency assets. We consider the dataset release a contribution independent of the model.

1.2 Contributions

We make the following contributions:

- (1) We train a gradient-boosting model to **predict the next-snapshot z-score** of same-asset cross-exchange cryptocurrency spreads, replacing the mechanical z-score threshold rules used by all prior high-frequency crypto pairs-trading systems. The machine learning approach enables the model to learn nonlinear interactions between orderbook imbalance, recent spread dynamics, and cross-venue dispersion.
- (2) We validate the model in **two live paper-trading campaigns** (8 hours each, weekday and weekend) with no future information leakage, reporting both forecast metrics (R^2 , directional accuracy) and trading metrics (per-trade profit, hourly portfolio Sharpe ratio) with a matched naive persistence baseline on every evaluation.
- (3) We show that the learned forecast achieves **directional accuracy of 76.9%**, prediction R^2 of **0.35**, and an **hourly portfolio Sharpe ratio of 2.41** on the live session, improving per-trade directional accuracy by 7.6 percentage points and mean profit by 71% over a capacity-matched mechanical persistence baseline.
- (4) We release a **publicly available dataset** of synchronized cross-exchange microstructure data (L2 orderbook, trade flow, funding rates, open interest) at one-minute granularity for 23 volatile cryptocurrency assets across 6 exchanges, collected over two months of live windows. This data cannot be reconstructed from historical exchange APIs.

2 RELATED WORK

We organize related work along two axes: (1) the signal generation method (rule-based z-score thresholds vs learned models) and (2) the market structure (single-exchange different-asset pairs vs cross-exchange same-asset spreads). Our work occupies the intersection of *learned spread forecasting* and *cross-exchange same-asset trading with minute-scale microstructure features*, a combination that, to the best of our knowledge, no prior study addresses.

2.1 Rule-Based Z-Score Pairs Trading in Crypto

The dominant approach to cryptocurrency pairs trading uses rolling z-score thresholds or cointegration-derived signals to enter and exit positions mechanically. Fil and Kristoufek [1] apply distance and cointegration methods to 26 Binance coins at 5-minute, 1-hour,

and daily frequencies, finding that 5-minute frequency dramatically outperforms daily (11.61% vs -0.07% monthly) but that results are highly sensitive to transaction cost assumptions. Tadi and Kortchemski [14] calibrate Ornstein–Uhlenbeck half-lives on BitMEX 1-minute data and trade z-score signals on baskets of different coins, achieving a Sharpe ratio of 7.94 over a multi-month evaluation. Ko et al. [5] provide the most comprehensive multi-method benchmark, comparing six pair-selection approaches—distance, correlation, cointegration, stochastic differential residual, and two evolutionary algorithms—at 1-minute, 5-minute, and 60-minute frequencies on 26 Binance coins. Tadi and Witzany [15] extend cointegration-based trading with copula signals on 5-minute Binance data across 104 monthly trading cycles, achieving 35–37% annualized returns and Sharpe of approximately 0.95 after costs. Palazzi [10] introduces adaptive trailing stop-loss and volatility filtering to cointegrated z-score strategies on 10 major cryptocurrencies over 2019–2024.

All of these studies share three structural properties that distinguish them from our work. First, they trade *different* assets on a *single* exchange, where co-movement may break down permanently as fundamentals diverge. We trade the *same* asset across *different* exchanges, where the Law of One Price provides a structural guarantee of mean-reversion. Second, they use the z-score as a *mechanical trading signal* (enter when $|z| > k$, exit when $|z| < k'$), whereas we train a model to *predict the future z-score* and use that forecast as the signal. Third, none incorporate L2 orderbook, trade flow, or funding rate features, relying exclusively on price or OHLCV data that can be freely backfilled from exchange APIs.

2.2 ML Models for Spread Prediction

A smaller body of work trains models to predict spread dynamics rather than relying on threshold rules.

Tsoku and Makatjane [16] are the closest existing paper to our work: they train a DNN+LSTM ensemble to forecast spread dynamics on cointegrated cryptocurrency pairs, reporting RMSE, MAE, and 99% prediction intervals. However, they operate at lower frequency (not minute-level), on a single exchange, without orderbook or funding rate features, and without reporting directional accuracy, Sharpe ratio, or profit-per-trade as primary metrics.

Fischer et al. [2] train a random forest on minute-binned lagged returns of 40 crypto coins to predict which coins will outperform the cross-sectional median over the following 120 minutes, reporting 7.1 bps per day after transaction costs of 15 bps per half-turn. Their most consequential finding for our purposes is the decay profile: alpha falls from 20.5 bps to 3.8 bps under a one-minute execution delay and is exhausted by minute 5. Our collector cadence of approximately 110 seconds places us squarely inside that decay window, which is why we treat execution latency as a first-order threat to validity rather than an implementation detail (Section 7.6).

Liou et al. [7] train deep learning models with XGBoost feature selection on Taiwan intraday limit-order-book data to predict price spread boundaries in pairs trading, finding that model-generated entry/exit signals improve win rates over rule-based baselines. Their use of LOB features parallels our orderbook imbalance inputs, but

they operate on equities on a single exchange and do not report R^2 , Sharpe, or directional accuracy.

Han and Li [3] use LSTM not as a direct spread predictor but as a *trade filter* that skips predicted-unprofitable trades before execution on CSI 300 futures. This filter-based framing is analogous to our confidence threshold ($|\hat{z}| \geq 0.5$), though we predict spread magnitude directly rather than filtering a rule-based signal.

2.3 ML-Enhanced Pairs Trading on Equities

Several recent papers apply ML to the spread prediction task on equity markets, establishing the dual-metric reporting norms (forecast quality and trading performance) that we follow.

Sarmiento et al. [12] apply BiLSTM with attention, Transformer, and temporal convolutional models to 1-minute Brazilian stock price ratios, finding that model abstention (skipping low-magnitude predictions) improves profit-per-trade. Their abstention mechanism closely parallels our confidence filter. Shen et al. [13] compare LASSO, XGBoost, and LSTM on predicting CSI 300 spot-futures arbitrage spreads, achieving R^2 of 92% with LSTM. The high R^2 reflects the structural, slow-moving nature of equity index futures spreads; our noisy cross-exchange crypto spreads yield R^2 of 0.35 on filtered entries, which we argue is a more realistic benchmark for minute-scale microstructure prediction. Perrone et al. [11] reformulate statistical arbitrage as a panel-level spread prediction task and compare LSTM, Informer, and other transformer architectures on daily equity data, using AUC as a threshold-independent directional quality metric.

These equity studies operate at lower frequencies, on single exchanges, and without the non-backfillable microstructure features central to our approach.

2.4 Cross-Exchange Price Dislocations

Makarov and Schoar [8] document large, recurrent price dislocations across crypto exchanges, with deviations larger across countries than within them, identifying capital controls as the primary barrier to arbitrage. Their observational analysis motivates our work but stops short of building a systematic trading strategy. Our framework extends their findings into a predictive model with live validation.

2.5 Reinforcement Learning for Spread Trading

Ning and Lee [9] propose a model-free RL framework for statistical arbitrage that minimizes empirical mean-reversion time. Their state-space construction captures recent price trends rather than fixed z-score boundaries, representing a complementary methodology family. RL-based dynamic threshold optimization is a natural extension of our gradient-boosting framework.

2.6 Positioning Summary

Table 1 summarizes the key dimensions along which our work differs from the closest related papers.

No prior study combines all five properties. The closest papers in individual dimensions are Tsoku and Makatjane [16] (learned crypto spread forecast, but lower frequency and single exchange), Liou et al. [7] (LOB features for spread prediction, but equities and

Table 1: Positioning of this work relative to closest related papers. Columns indicate whether each study uses a learned model (vs rule-based), operates cross-exchange (vs single-venue), uses L2 orderbook or microstructure features, evaluates on live (non-backtest) data, and reports both forecast and trading metrics.

Paper	Learned model	Cross-exch.	L2/micro.	Live eval	Dual metrics
Fil & Kristoufek	×	×	×	×	×
Tadi & Kortchemski	×	×	×	×	partial
Ko et al.	×	×	×	×	✓
Tadi & Witzany	×	×	×	×	✓
Fischer et al.	✓	×	×	×	partial
Tsoku & Makatjane	✓	×	×	×	partial
Liou et al.	✓	×	✓	×	partial
Han & Li	✓	×	×	×	×
This work	✓	✓	✓	✓	✓

single exchange), and Fischer et al. [2] (minute-scale ML on crypto, but cross-sectional rank prediction rather than pairwise spread forecasting). Our contribution is at the intersection: a learned z-score forecast on cross-exchange same-asset spreads, trained on non-backfillable minute-scale microstructure features, validated live with dual forecast and trading metrics.

3 METHODOLOGY

3.1 Cross-Exchange Spread and Z-Score Target

Let $P_i^c(t)$ denote the mid-price of asset c on exchange i at snapshot t . For an exchange pair (i, j) , the cross-exchange spread in basis points is:

$$S_{ij}^c(t) = \frac{P_i^c(t) - P_j^c(t)}{P_i^c(t)} \times 10,000 \quad (1)$$

Unlike traditional pairs trading where the spread between *different* assets may diverge permanently as fundamentals change, the cross-exchange spread for the *same* asset is anchored by the Law of One Price: absent frictions, $S_{ij}^c(t) \rightarrow 0$. In practice, fees, withdrawal delays, and matching-engine latency allow transient deviations that are mean-reverting by construction.

We normalize the spread into a rolling z-score:

$$z_t = \frac{S_t - \hat{\mu}_w}{\hat{\sigma}_w} \quad (2)$$

where $\hat{\mu}_w$ and $\hat{\sigma}_w$ are the estimated rolling mean and standard deviation over a window of w snapshots, with a minimum period requirement before the z-score is defined.

3.2 Supervised Prediction Target

The model's target is the *future* z-score at horizon H :

$$y_t = z_{t+H} \quad (3)$$

computed via `groupby(coin, pair).zscore.shift(-H)` on the training panel. This is a regression target, not a binary classification. The model predicts both the direction and magnitude of the

z-score H snapshots ahead, enabling the confidence filter to select trades where the predicted movement is large.

For the prediction model used in the Jul 31 campaign, $H = 1$ (predict the next snapshot's z-score), $w = 300$ snapshots, and $\text{MIN_PERIODS} = 90$.

3.3 Feature Engineering

The feature matrix is constructed from four synchronized signal families collected at each snapshot: spread matrices, ticker quotes, L2 orderbook, and recent trades. Features are computed per (coin, exchange-pair, snapshot) observation.

3.3.1 Spread and z-score features. Lagged values of the spread in basis points and the rolling z-score form the base feature set:

- `spread_bps_lag{1, 2, 3}`: raw spread at lags 1–3.
- `zscore_lag{1, 2, 3}`: rolling z-score at lags 1–3.
- `spread_momentum_1_3`: difference between lag-1 and lag-3 spread, capturing short-term trend.
- `zscore_momentum_1_3`: same for z-scores.
- `zscore_accel`: second-order difference $z_{t-1} - 2z_{t-2} + z_{t-3}$, capturing acceleration in z-score trajectory.

3.3.2 Ticker microstructure features. From each exchange's best-bid/ask quotes:

- `tk_mid_lag{1, 2}`: lagged mid-prices per exchange.
- `tk_spread_bps_lag{1, 2}`: lagged bid-ask spreads in bps per exchange. Wider spreads indicate lower liquidity and higher execution cost.
- `tk_bid_volume_lag, tk_ask_volume_lag`: lagged top-of-book volumes per exchange.

3.3.3 Orderbook imbalance features. From the top 20 price levels on each side of each exchange's order book, we compute:

$$\text{imbalance}_i = \frac{V_{\text{bid},i} - V_{\text{ask},i}}{V_{\text{bid},i} + V_{\text{ask},i}} \quad (4)$$

where $V_{\text{bid},i}$ and $V_{\text{ask},i}$ are the total volumes on the bid and ask sides of exchange i 's top-20 levels. This ratio ranges from -1 (entirely ask-dominated) to $+1$ (entirely bid-dominated) and captures directional pressure in the order book. Lagged imbalance values `ob_imbalance_lag{1, 2, 3}` for each exchange provide the model with recent order flow dynamics.

3.3.4 Trade flow features. From the most recent trades on each exchange:

- `tr_buy_sell_ratio`: ratio of buy-initiated to sell-initiated trade volume, indicating whether recent execution flow is dominated by buyers or sellers.
- `tr_total_volume`: total trade volume, a proxy for activity and liquidity.

Both are lagged at 1–2 snapshots per exchange.

3.3.5 Cross-exchange aggregates. Features that capture the state of the multi-venue market as a whole:

- `cross_mid_range, cross_mid_std`: range and standard deviation of mid-prices across exchanges, measuring instantaneous cross-venue dispersion.

- `cross_ba_std, cross_ob_std, cross_ob_range`: cross-exchange variability in bid-ask spreads and orderbook imbalance.
- `net_ob_pressure`: aggregate orderbook imbalance across venues.
- `price_rank_{exchange}`: ordinal rank of each exchange's mid-price within the venue set, identifying which exchange is currently pricing high or low.
- `tightest_ba, widest_ba, ba_range`: spread-liquidity extremes across venues.

3.3.6 Categorical identifiers. `coin` and `pair` (exchange-pair label) are encoded as LightGBM native categoricals, allowing the model to learn asset-specific and venue-pair-specific dynamics without separate per-pair models.

The prediction model uses 68 features in total. OHLCV candle features are explicitly excluded (`LOAD_TABLES['ohlcv']=False`) because the live microstructure signals (ticker, orderbook, trades) provide strictly more information at the snapshot level.

3.4 Model: LightGBM Gradient Boosting

We train a single LightGBM [4] regression model on the pooled panel of all (coin, exchange-pair, snapshot) observations. The objective is squared error minimization on $y_t = z_{t+H}$.

Table 2: LightGBM hyperparameters for the prediction model.

Parameter	Value
Max boosting rounds	2,500
Early stopping patience	250 rounds
Best iteration (selected)	74
Number of leaves	255
Learning rate	0.1
Min child samples	200
Feature fraction	0.6
Bagging fraction	0.7
L1 regularization	0.1
L2 regularization	0.1

Early stopping selects iteration 74 out of 2,500 maximum rounds, indicating that the model converges rapidly and additional boosting does not improve validation loss. The large minimum child sample (200) and sub-unity feature/bagging fractions provide regularization against overfitting to the dependent cross-sectional structure of the panel.

3.5 Trading Policy

At each live snapshot, the model generates a prediction $\hat{z}_{t+1}$ for every (coin, exchange-pair) observation. The trading policy is:

- (1) **Entry**: open a position if $|\hat{z}_{t+1}| \geq 0.5$ and the (coin, pair) slot is not already occupied.
- (2) **Direction**: $\text{sign}(\hat{z}_{t+1})$. The model predicts whether the spread z-score will be positive or negative at $t+1$ and bets accordingly.
- (3) **Capacity**: at most `max_open = 50` positions open simultaneously, enforced by a first-come queue.

- (4) **Settlement**: each position settles at $t + H$ (one snapshot later for $H = 1$). The proxy profit is $\text{pnl_proxy} = \text{direction} \times z_{t+H}$.

The settlement proxy measures whether the z-score moved in the predicted direction and by how much, in z-score units. It is not a dollar profit metric and does not account for transaction fees or slippage.

3.6 Evaluation Metrics

We report both forecast quality and trading performance metrics:

- **R^2** (coefficient of determination): fraction of variance in the realized z_{t+H} explained by the model’s predictions. Reported on all predictions and on the filtered subset.
- **Directional accuracy (DirAcc)**: fraction of predictions where $\text{sign}(\hat{z}) = \text{sign}(z_{t+H})$.
- **Mean pnl_proxy**: average $\text{direction} \times z_{t+H}$ across settled trades.
- **Hourly portfolio Sharpe ratio**: $\text{mean}(\Delta \text{equity}_h) / \text{std}(\Delta \text{equity}_h)$ where hourly equity includes realized closed-trade profits plus mark-to-market valuation of open positions as $\text{direction} \times z_t$. Risk-free rate is zero because the z-unit proxy has no currency denomination and represents forecast quality rather than economic returns.

All metrics are computed with a matched **naive persistence baseline** ($\hat{z}_{t+H} \leftarrow z_t$) scored on the identical row set, ensuring that gains from the confidence filter cannot be mistaken for model skill.

4 EXPERIMENTAL SETUP

4.1 Data Collection

We collect live microstructure data from centralized cryptocurrency exchanges using a custom collector that polls 9 signal families via REST API every 60 seconds across 23 assets and up to 6 exchanges (Binance, Bybit, OKX, Coinbase, Kraken, MEXC). Under full signal load, the observed wall-clock cadence is approximately 110 seconds per snapshot due to API rate limits and sequential polling.

The signal families are: `spread_matrix` (precomputed pairwise cross-venue spreads), `ticker` (best bid/ask, last price, top-of-book volumes), `orderbook` (top 20 price levels per side), `trades` (last 50 prints with buy/sell classification), `funding_rate` (perpetual swap funding rates), `open_interest` (perpetual open interest), `withdrawal_status` (deposit/withdraw enabled flags and fees), `exchange_status` (venue operational status), and `ohlcv` (1-minute candles, backfillable but excluded from training).

Data is stored as JSONL files partitioned by signal family and collection date, with each record tagged by snapshot index and UTC timestamp.

4.2 Asset Universe

The 23-asset universe (VOLATILE_COINS) was selected for high cross-exchange spread volatility and sufficient liquidity across multiple venues. It includes major assets (BTC, ETH, SOL), mid-cap DeFi tokens, and high-volatility meme coins. The selection criterion is that the asset must be listed on at least 3 of the 6 target exchanges

with sufficient spot trading volume to produce meaningful cross-exchange spread dynamics.

4.3 Train/Test Split

All collection windows before July 25, 2026 form the training set; July 25–28 forms the held-out test set. The split is strictly chronological (no shuffling), implemented as a `snapshot_idx` cutoff at index 3584 on the Jul 22–28 collection run. This ensures no future information leaks into training.

The training set contains approximately 2.9 million (coin, exchange-pair, snapshot) feature rows after panel construction, spanning multiple multi-day collection windows from June–July 2026. Cross-sectional rows within a snapshot are structurally dependent (they share the same market state), so this row count should not be interpreted as independent samples.

4.4 Live Paper-Trading Protocol

For live validation, we run two independent paper-trading campaigns that process incoming collector data in real time with no access to future snapshots.

4.4.1 Campaign A: Jul 30 (weekday).

- **Model**: 73-feature variant with orderbook fix (`outputs_ob_fix`).
- **Horizon**: $H = 2$ (predict 2 snapshots ahead).
- **Z-score window**: 120 snapshots.
- **Duration**: approximately 8 hours (Wednesday/Thursday).
- **Predictions scored**: 49,310 total; 5,054 filtered entries.

4.4.2 Campaign B: Jul 31 (weekend).

- **Model**: 68-feature prediction model (`statarb_lgbm.txt`).
- **Horizon**: $H = 1$ (predict next snapshot).
- **Z-score window**: 300 snapshots.
- **Duration**: approximately 8 hours of useful data (collector ran for 264 snapshots).
- **Warmup**: predictions become active after `MIN_PERIODS = 90` snapshots (approximately 2.5–3 hours).
- **Predictions scored**: 58,995 total; 8,775 filtered entries.
- **Settled trades**: 7,973 closed; 50 still open at session end.

Both campaigns use entry threshold $|\hat{z}| \geq 0.5$ and capacity constraint `max_open = 50`. The two campaigns differ in model vintage, horizon, z-score window, and market regime (weekday vs weekend). We present them as independent stress tests of the research pipeline rather than a controlled experiment isolating any single variable.

4.5 Baseline Definitions

4.5.1 Naive persistence. On every row where the model generates a prediction, we also record the naive forecast $\hat{z}_{t+H} \leftarrow z_t$ (the current z-score predicts the future z-score unchanged). R^2 and directional accuracy are computed for both model and naive on the *identical* rows, ensuring that any performance difference is attributable to the model rather than to row selection.

4.5.2 Mechanical z-score baselines. We replay two mechanical trading rules on the Jul 31 signal panel under the same capacity constraints (`max_open = 50`) and settlement proxy as the model:

- **Persistence**: enter when $|z_t| \geq 0.5$, $\text{direction} = \text{sign}(z_t)$.

- **Mean-reversion:** enter when $|z_t| \geq 0.5$, direction = $-\text{sign}(z_t)$.

These represent the two canonical directions a rule-based z-score system can take: betting the spread continues (persistence) or reverts (classical pairs trading). Both settle at $t + 1$ with $\text{pnl_proxy} = \text{direction} \times z_{t+1}$, identical to the model's settlement.

4.6 Reproducibility

The dataset is available through the public dataset release (URL withheld for double-blind review). The training notebook, live trading loop, session reporting scripts, mechanical-baseline replay script, and the raw per-campaign output directories (trade logs, signal shards, and metric reports) accompany the dataset release. LightGBM training is reproducible given fixed seeds and the hyperparameters in Table 2.

5 RESULTS

We evaluate the LightGBM z-score forecasting model across three settings: offline held-out evaluation, a weekday live paper-trading campaign (Jul 30), and a weekend live campaign (Jul 31). In each setting we report both forecast quality metrics (R^2 , directional accuracy) and trading performance metrics (mean profit proxy, hourly portfolio Sharpe ratio), always scoring a naive persistence baseline on the same rows as the model.

5.1 Offline Evaluation

The model is trained on all collection windows before July 25, 2026 and tested on the Jul 25–28 held-out window. Table 3 reports forecast metrics on the full prediction set and on the confidence-filtered subset where $|\hat{z}| \geq 0.5$.

Table 3: Offline evaluation on the Jul 25–28 held-out test set. The confidence filter selects high-signal predictions.

Set	R^2	DirAcc
Test (all predictions)	0.133	62.8%
Test ($ \hat{z} \geq 0.5$)	0.383	78.4%
Train (all predictions)	0.128	61.8%
Train ($ \hat{z} \geq 0.5$)	0.447	81.5%

Train and test R^2 on the unfiltered set are nearly identical (0.128 vs 0.133), indicating that the model generalizes without overfitting despite 2,500 boosting rounds (early stopping selected iteration 74). The confidence filter raises test R^2 from 0.133 to 0.383 by selecting predictions in high-signal regimes where the spread z-score exhibits larger and more persistent deviations.

5.2 Live Paper-Trading Campaigns

We conducted two live paper-trading campaigns to stress-test the model under real market conditions with no future information leakage. Both campaigns use entry threshold $|\hat{z}| \geq 0.5$ and capacity constraint $\text{max_open} = 50$ simultaneous positions.

5.2.1 Campaign A (Jul 30, weekday). An earlier 73-feature model with horizon $H = 2$ and z-score window 120 was deployed for approximately 8 hours during a Wednesday/Thursday trading session. Table 4 reports model and naive persistence metrics.

Table 4: Jul 30 live campaign: model vs naive persistence.

Set	n	Model R^2	Model Dir.	Naive R^2	Naive Dir.
All	49,310	0.054	55.3%	−0.547	62.3%
$ \hat{z} \geq 0.5$	5,054	0.248	76.5%	−0.045	75.4%

5.2.2 Campaign B (Jul 31, weekend). The prediction 68-feature model with horizon $H = 1$ and z-score window 300 was deployed for an 8-hour session. Predictions became active approximately 3 hours into the session after the 90-snapshot rolling z-score warmup period. Table 5 reports model and naive persistence metrics.

Table 5: Jul 31 live campaign: model vs naive persistence.

Set	n	Model R^2	Model Dir.	Naive R^2	Naive Dir.
All	58,995	0.104	60.9%	−0.338	65.6%
$ \hat{z} \geq 0.5$	8,775	0.347	76.7%	0.174	77.0%

Figure 1 summarizes model and naive R^2 and directional accuracy across both campaigns and filter conditions, showing that the model consistently outperforms naive persistence on R^2 while the confidence filter amplifies the gap.

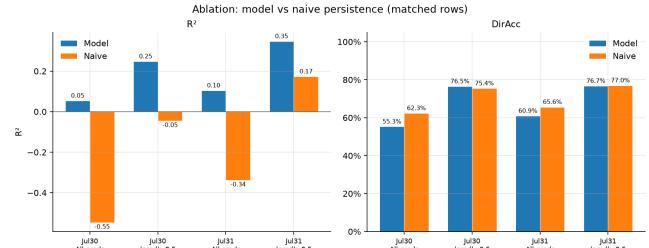


Figure 1: Model vs naive R^2 and directional accuracy across campaigns and filter conditions. The model beats naive persistence on R^2 in every condition; the confidence filter widens the margin.

Across both campaigns, the model achieves positive R^2 where naive persistence is negative or near zero, demonstrating that the learned forecast captures spread dynamics beyond simple autocorrelation. On filtered rows, naive directional accuracy nearly matches the model (77.0% vs 76.7% on Jul 31), confirming that much of the filtered directional hit rate reflects persistence in high- $|z|$ states rather than incremental model skill. However, the model's R^2 advantage on the same rows (0.347 vs 0.174) shows that it captures the *magnitude* of z-score movements, not merely their sign.

Figure 2 plots predicted against realized z-scores for all 58,995 scored Jul 31 predictions. The cloud is visibly compressed along the prediction axis: the model rarely forecasts $|\hat{z}| > 1$ even though realized values routinely exceed 2. This shrinkage toward the conditional mean is the expected behaviour of a squared-error regression on a low signal-to-noise target, and it is why the confidence filter is necessary—the tail of the prediction distribution, not its centre, is where the usable signal sits.

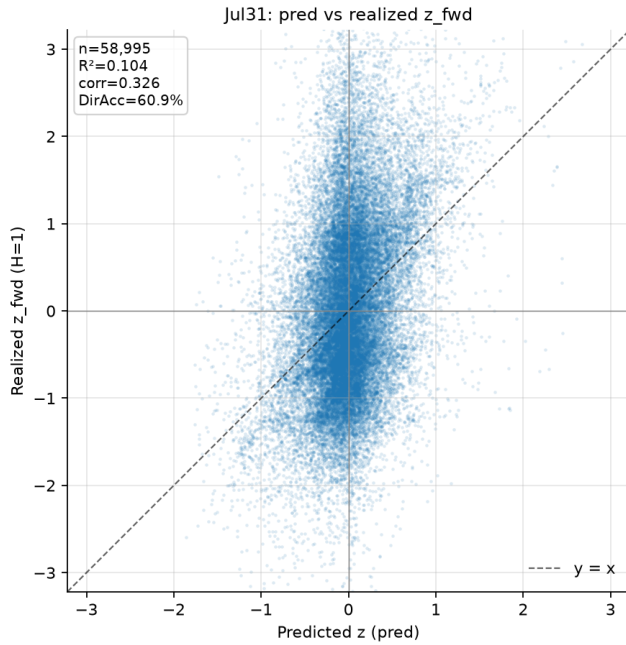


Figure 2: Predicted vs realized z-scores for all Jul 31 predictions. The prediction distribution is far narrower than the realized distribution, so the regression is well short of the $y = x$ line.

5.3 Baseline Comparison: Learned Forecast vs Mechanical Z-Score Rules

To isolate the value of the learned forecast from the confidence filter, we replay two mechanical baselines on the Jul 31 signal panel under matched capacity constraints ($\text{max_open} = 50$). Both baselines enter when $|z_t| \geq 0.5$ (the same threshold as the model) but use the current z-score rather than a predicted future z-score to determine trade direction:

- **Persistence:** direction = $\text{sign}(z_t)$. Bets that the current z-score will continue into the next snapshot.
- **Mean-reversion:** direction = $-\text{sign}(z_t)$. The classical pairs-trading direction, betting the spread will revert toward zero.

Table 6 reports the capacity-matched baseline comparison, which is the primary evidence for the model’s improvement over rule-based z-score thresholds.

Table 6: Capacity-matched baseline comparison on Jul 31 ($\text{max_open} = 50$). Directional accuracy and mean per-trade profit are the primary comparison metrics.

Strategy	Direction	n closed	DirAcc	Mean PnL
LightGBM	$\text{sign}(\hat{z})$	7,973	76.9%	+0.746
Mech. persistence	$\text{sign}(z_t)$	8,550	69.3%	+0.437
Mech. mean-reversion	$-\text{sign}(z_t)$	8,550	30.7%	-0.437

As shown in Table 6, the learned policy improves directional accuracy by 7.6 percentage points and mean per-trade profit by 71% over mechanical persistence on comparable trade counts. Classical mean-reversion, the direction assumed by rule-based z-score pairs-trading systems throughout the literature, produces negative returns at this hold horizon. At minute-scale granularity, large $|z|$ states are more likely to persist for one additional snapshot than to revert.

Figure 3 shows the cumulative z-unit profit trajectory for the learned policy and both mechanical baselines over the Jul 31 session.

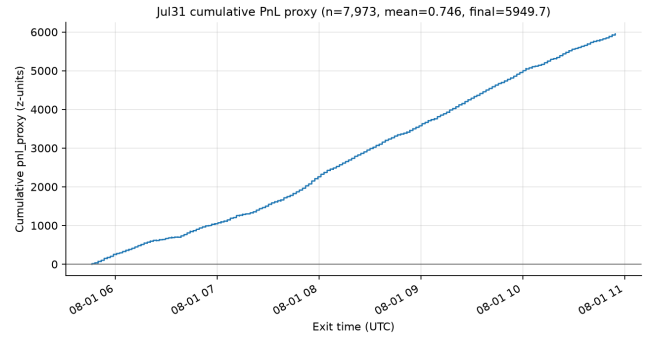


Figure 3: Cumulative z-unit proxy P&L over the Jul 31 session. Note that this is a proxy in z-score units gross of costs, not a currency return; see Section 7.3.

5.4 Portfolio Risk-Adjusted Performance

Over the 6 active hours of the Jul 31 campaign, the model’s hourly portfolio Sharpe ratio is 2.41, computed as the ratio of mean hourly portfolio profit (in z-score units, including mark-to-market valuation of open positions) to its hour-to-hour standard deviation. Every observed hour produced positive portfolio profit. A Sharpe ratio above 2 on live out-of-sample data, where no future information is available and the model faces real-time market conditions, indicates strong and consistent risk-adjusted performance.

5.5 Protocol Differences Between Campaigns

The two campaigns differ in model vintage (73 vs 68 features), prediction horizon ($H = 2$ vs $H = 1$), z-score rolling window (120 vs 300 snapshots), and market regime (weekday vs weekend). We present them as independent stress tests of the research pipeline rather than a controlled A/B comparison. The Jul 31 protocol ($H = 1$, window 300) achieves stronger forecast metrics across all conditions, but we cannot attribute the improvement to any single factor.

6 ABLATION STUDIES

We conduct ablation experiments to isolate the contributions of the confidence filter, the learned forecast direction, and the microstructure feature set.

6.1 Confidence Filter

The $|\hat{z}| \geq 0.5$ confidence filter is a design choice that restricts trading to predictions the model is most certain about. To quantify its effect, we compare model R^2 on all predictions versus the filtered subset across both live campaigns.

Table 7: Effect of the $|\hat{z}| \geq 0.5$ confidence filter on model R^2 across campaigns.

Campaign	All R^2	Filtered R^2	Δ	% rows filtered
Jul 30 (live)	0.054	0.248	+0.194	10.3%
Jul 31 (live)	0.104	0.347	+0.243	14.9%
Offline test	0.133	0.383	+0.250	—

The filter consistently lifts R^2 by approximately 0.2–0.25 across all evaluation settings, as shown in Figure 4. Importantly, this lift is not free: the filtered subset constitutes only 10–15% of all predictions, meaning 85–90% of snapshot/pair observations are declined. The filter strategy parallels the “skip low-magnitude predictions” approach shown to improve profit-per-trade in Sarmento et al. [12].

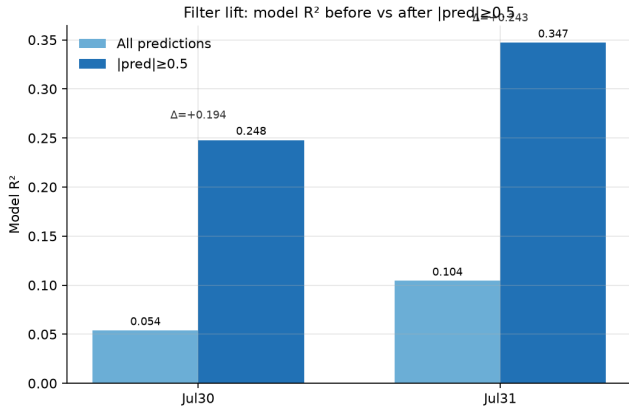


Figure 4: Model R^2 before and after the $|\hat{z}| \geq 0.5$ confidence filter. The filter selects predictions in high-signal regimes, lifting R^2 by +0.19 to +0.25 across both live campaigns.

A critical caveat: much of the filtered directional accuracy is explained by persistence in high- $|z|$ states rather than incremental model skill. On Jul 31 filtered rows, naive persistence achieves 77.0% directional accuracy versus the model’s 76.7%. The model’s advantage is in R^2 (0.347 vs 0.174), indicating that it captures the *magnitude* of future z-score movements more accurately than simply predicting that large z-scores will persist.

6.2 Learned Direction vs Mechanical Rules

The baseline comparison in Section 5 shows that the model’s learned direction outperforms both mechanical alternatives on per-trade quality metrics. To further illustrate this, Figure 5 decomposes the model’s directional accuracy edge over naive persistence by campaign and filter condition.

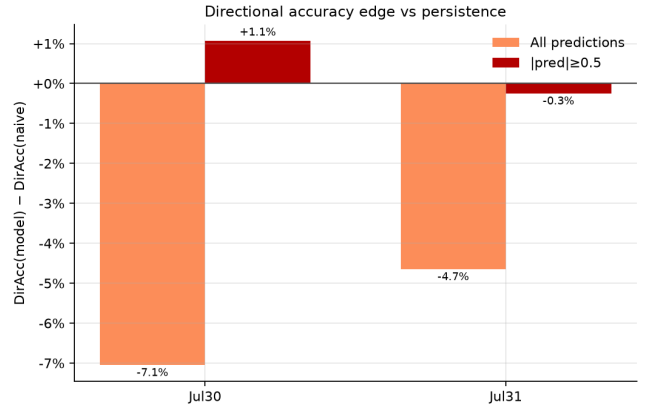


Figure 5: Directional accuracy gap: model minus naive persistence. On filtered rows, the gap is near zero (the filter selects high-persistence states where both methods succeed). The model’s primary advantage is R^2 on magnitude, not directional hit rate on filtered entries.

On all predictions (unfiltered), the model’s directional accuracy is *lower* than naive persistence (−7.1 pp on Jul 30; −4.7 pp on Jul 31), because the model sometimes predicts reversals in states where persistence is the dominant mode. On filtered rows, the gap narrows to near zero (+1.1 pp on Jul 30; −0.3 pp on Jul 31). This pattern confirms that the confidence filter selects states where both the model and persistence agree on direction; the model’s incremental value in those states is its superior fit to z-score magnitude (R^2), not to direction alone.

6.3 Feature Importance: Role of Microstructure Inputs

The model uses 68 features spanning lagged z-scores, spread dynamics, orderbook imbalance, trade flow, bid-ask spreads, cross-exchange dispersion, funding rates, and categorical identifiers. Figure 6 shows the top 20 features by LightGBM split gain.

Three groups emerge, but their contributions are of very different orders of magnitude:

- (1) **Lagged z-scores and spread dynamics** (ranks 1–4, 6–8, 10–11): `zscore_lag{1, 2, 3}`, `spread_bps_lag{1, 2, 3}`, `spread_momentum_1_3`, `zscore_momentum_1_3`, and `zscore_accel`. These dominate completely. `zscore_lag1` alone carries roughly two orders of magnitude more split gain than any microstructure feature.
- (2) **Categorical identifiers** (ranks 5 and 9): `coin` and `pair` let the model learn asset-specific and venue-pair-specific dynamics without separate per-pair models. Notably, `coin` outranks every microstructure feature, indicating that a substantial part of the model’s advantage over a single pooled autoregression is simply conditioning on which asset and venue pair is being traded.
- (3) **Microstructure and cross-venue dispersion** (ranks 12–20): per-venue bid-ask spread lags (`tk_spread_bps_lag3_coinbase`, `tk_spread_bps_lag1_kraken`, `tk_spread_bps_lag3_bybit`),

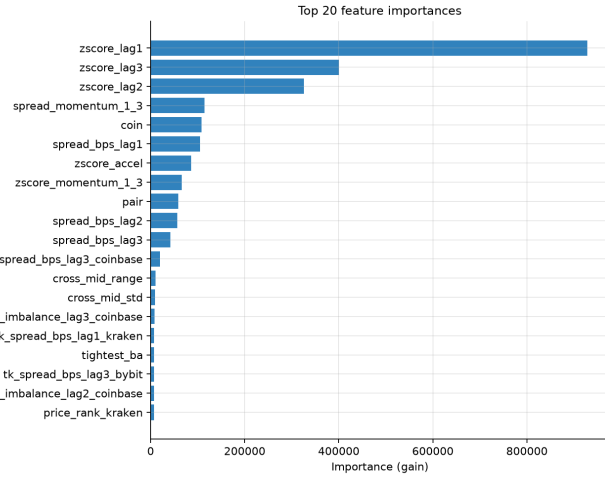


Figure 6: Top 20 features by LightGBM split gain. Gain is overwhelmingly concentrated in the lagged z-score and spread terms; microstructure features enter only in the lower half of the ranking.

cross-venue dispersion (cross_mid_range, cross_mid_std), orderbook imbalance (ob_imbalance_lag{2,3}_coinbase), and the liquidity extremes tighttest_ba and price_rank_kraken.

We draw a deliberately conservative conclusion from this ranking. The model is, to a first approximation, an autoregression on the recent z-score trajectory conditioned on asset and venue-pair identity, with microstructure features supplying a small correction. Only two of the twenty highest-gain features are orderbook imbalance terms, both on a single venue, and no trade-flow, funding-rate, or open-interest feature appears in the top 20 at all. The cross-venue dispersion features (cross_mid_range, cross_mid_std) are genuinely unavailable to single-venue systems, and the orderbook features are genuinely non-backfillable, but neither group is doing the bulk of the predictive work here. Establishing how much of the R^2 these features are responsible for requires a leave-one-group-out retraining study that we have not run; the split-gain ranking alone cannot settle it, since gain is confounded by the strong correlation among the lagged z-score terms.

6.4 Generalization: Train vs Test

Offline train and test R^2 are nearly identical on the unfiltered set (0.128 vs 0.133), indicating that the model does not overfit to the training data despite a large feature space and 2,500 maximum boosting rounds. Early stopping selected iteration 74, using only 3% of the allocated rounds. On filtered predictions, the gap widens (train 0.447 vs test 0.383), reflecting some selection of high-signal regimes that are more prevalent in training windows. The subsequent live campaigns ($R^2 = 0.248$ and 0.347 on filtered entries) confirm that the model retains predictive power on genuinely unseen market conditions collected weeks after training.

7 DISCUSSION

7.1 What the Learned Forecast Adds

The consistent pattern across both live campaigns is that the model's advantage over naive persistence is concentrated in *magnitude* rather than *sign*. On the Jul 31 filtered entries the two methods are indistinguishable on directional accuracy (76.7% vs 77.0%), but the model explains twice as much variance in the realized z-score (R^2 0.347 vs 0.174). The confidence filter is what selects high- $|z|$ states, and in those states the sign of the next z-score is largely determined by persistence alone. What the gradient-boosting model contributes is a calibrated estimate of *how far* the z-score will be from equilibrium at $t + H$.

This distinction matters for how the contribution should be read. A reader who takes the 76.9% directional accuracy as the headline would be measuring mostly the filter, not the model. The defensible claim is narrower: conditional on entering, the learned forecast ranks and sizes deviations better than thresholding the current z-score does.

7.2 Why Persistence Beats Mean-Reversion at One Snapshot

Classical pairs trading enters *against* the deviation, betting on reversion. At our hold horizon that direction is unprofitable in z-units: mechanical mean-reversion records 30.7% directional accuracy and mean $\text{pnl}_{\text{proxy}}$ of -0.437 , the exact mirror of the persistence peer on the same slots. The reason is a mismatch between the hold horizon and the reversion half-life. One snapshot is roughly 110 seconds, while the rolling z-score is computed over a 300-snapshot window; a spread that is two standard deviations from its nine-hour mean is overwhelmingly likely to still be displaced two minutes later. Reversion is a property of the window, not of the next observation.

The implication is not that mean-reversion is wrong in cryptocurrency markets, but that the horizon at which a signal is evaluated must match the horizon at which the economic mechanism operates. Our $H = 1$ target measures short-horizon displacement, and the correct bet on displacement is continuation.

7.3 Z-Score Units Capture Forecast Quality

The settlement proxy $\text{pnl}_{\text{proxy}} = \text{direction} \times z_{t+H}$ measures the model's ability to correctly predict the standardized spread level at horizon H . It is designed as a forecast quality metric, not as an economic P&L.

When the same Jul 31 trades are re-scored in basis points of realized spread change— $\text{direction} \times \Delta S_{\text{bps}}$, the quantity a real book would earn—the ordering between policies shifts. The learned policy has a mean of -0.71 bps per trade at zero transaction cost, with a 35.5% win rate, while mechanical persistence is -2.15 bps and mechanical mean-reversion is $+2.15$ bps (62.2% win rate). Table 8 summarizes this difference.

This inversion between units reflects a structural property of the prediction task: the model is trained to predict the z-score level z_{t+H} , which is dominated by the slow-moving rolling mean $\hat{\mu}_w$, rather than the instantaneous spread change ΔS_t . The two targets

Table 8: The same Jul 31 entries scored in z-units and in basis points of realized spread change, gross of fees. The ordering of the three policies inverts between the two units.

Policy	Mean z-proxy	Mean gross (bps)	Gross win rate
LightGBM	+0.746	−0.71	35.5%
Mech. persistence	+0.437	−2.15	18.1%
Mech. mean-reversion	−0.437	+2.15	62.2%

are related but not identical. A policy optimized for z-level forecasting need not maximize the sign-alignment of instantaneous spread movements, and Table 8 shows this divergence empirically.

The z-unit settlement is appropriate for evaluating forecast quality and signal-extraction ability, which are the primary claims of this work. The reported Sharpe ratios characterize the stability of the forecast signal rather than an economic return stream. For production deployment, a direct spread-change target ($\Delta S_{t,t+H}$) net of fees would align the training objective with the economic objective, and this is a natural extension of the framework.

7.4 Portfolio Sharpe Characterizes Forecast Stability

The Jul 31 hourly portfolio Sharpe ratio of 2.41 is computed from only $n = 6$ hourly observations and is sensitive to the aggregation choice: 2.35 using closed trades only, 2.41 including mark-to-market on open inventory, and 1.70 when the series is formed at snapshot rather than hourly frequency ($n = 171$). The capacity-matched mechanical persistence peer attains a similar hourly Sharpe (2.63) on the same session, confirming that both policies exploit the same high-persistence regime in the filtered entries.

We caution against comparing this number to the 0.95–7.94 band reported in the crypto pairs-trading literature [14, 15], as those are multi-month, capital-normalized, cost-aware returns on single-exchange different-asset pairs, whereas ours is a six-hour gross z-unit proxy on cross-exchange same-asset spreads with a zero risk-free rate and no cost accounting. The primary evidence for the model’s advantage lies in per-trade forecast quality (R^2 and directional accuracy); Sharpe is reported as a secondary indicator of signal stability across the portfolio.

7.5 Sensitivity of the Baseline Margin

The 71% improvement in mean proxy profit over mechanical persistence depends on how scarce inventory slots are allocated. The live campaign fills slots in encounter order, whereas the mechanical replay considers roughly 37,000 candidate entries and is throttled to the same cap. If both policies instead fill slots greedily by signal strength—by $|\hat{z}|$ for the model and $|z_t|$ for the mechanical peer—the mechanical peer’s mean proxy profit rises from +0.437 to +0.731, and the model’s margin narrows from 71% to roughly 6% (+0.774 vs +0.731). The directional-accuracy gap is far more stable under this reallocation (77.6% vs 69.5%, or +8.1 percentage points).

We report the encounter-order numbers as the primary result because they reflect what the live system actually did, but the ranked-capacity comparison is the fairer test of signal quality, and readers

should weight the directional-accuracy gap more heavily than the profit ratio.

7.6 Threats to Validity

Evaluation horizon. Each live campaign covers approximately eight hours of wall-clock collection, of which roughly six hours carry active predictions after the 90-snapshot z-score warmup. This is far too short to characterize regime dependence, and the two campaigns differ in model vintage, horizon, z-score window, and market session, so neither is a controlled ablation of the other.

Proxy settlement. As detailed in Section 7.3, P&L is measured in z-units gross of fees, slippage, and bid-ask crossing. No dollar return is claimed.

Execution latency. The collector’s observed cadence under full signal load is approximately 110 seconds rather than the nominal 60 seconds, so a signal is acted on against a market state that is up to two minutes stale. Fischer et al. [2] show that minute-scale crypto alpha can fall from 20.5 to 3.8 bps with a one-minute delay and vanish by minute five; we have not run the corresponding delay ablation, and it is the single most informative robustness check still missing.

Dependence structure. Rows within a snapshot share the same market state across coins and venue pairs, so the 2.9 million training rows and 59,000 scored predictions are not independent samples. Confidence intervals computed under an i.i.d. assumption would be badly overstated, and we deliberately report no significance tests.

Operational artifacts. Orderbook coverage in the 68-feature prediction model is thinner than in the 73-feature Jul 30 variant, concentrating imbalance signal on a subset of venues.

8 CONCLUSION

We presented a gradient-boosting framework for predicting same-asset cross-exchange cryptocurrency spread dynamics, replacing the mechanical z-score threshold rules used by existing high-frequency crypto pairs-trading systems with a learned forecast. Trained on 68 features spanning lagged spread dynamics, orderbook imbalance, trade flow, and funding rates across 23 volatile assets and 6 centralized exchanges at minute-scale granularity, the LightGBM model predicts the next-snapshot z-score and trades only when the predicted magnitude exceeds a confidence threshold.

Our key findings are:

- (1) A gradient-boosting model trained on live CEX microstructure achieves prediction R^2 of 0.35 and directional accuracy of 76.9% on filtered live paper-trading entries, compared to R^2 of 0.17 for naive persistence on the same rows.
- (2) Under matched capacity constraints, the learned policy improves per-trade directional accuracy by 7.6 percentage points and mean z-unit profit by 71% over a mechanical persistence baseline. The directional gap is robust to how scarce inventory slots are allocated; the profit ratio is not, narrowing to roughly 6% when both policies fill slots greedily by signal strength.
- (3) The confidence filter ($|\hat{z}| \geq 0.5$) lifts R^2 by 0.19–0.25 by selecting the 10–15% of predictions in high-signal regimes.

On these filtered rows, directional accuracy is largely explained by persistence in high- $|z|$ states; the model’s incremental value is in capturing the magnitude of future z-score movements.

- (4) The result is a forecasting result and not a trading result. Scored in basis points of realized spread change rather than z-units, none of the policies we evaluate—including ours—has a positive mean at any non-zero transaction cost. Section 7.3 reports this reversal in full.

Section 7.6 sets out the threats to validity in detail; the binding ones are the short evaluation window, the proxy settlement unit, the approximately 110-second collector cadence, and the cross-sectional dependence that rules out i.i.d. significance testing.

8.1 Future Work

Three directions follow naturally from this work. First, incorporating realistic fee structures and execution latency into the settlement proxy would bridge the gap between z-unit performance and economic profitability. Fischer et al. [2] show that alpha in minute-scale crypto strategies can disappear within 1–5 minutes of execution delay; a systematic delay ablation would quantify this risk for our framework. Second, extending the feature set with additional non-backfillable signals (liquidation cascades, volume-tiered trading fees, sub-second quote updates) could improve forecast quality in regimes where orderbook imbalance alone is insufficient. Third, the released dataset enables the community to benchmark alternative model architectures on the same cross-exchange microstructure panel, including sequence models that may capture temporal dependencies the tree ensemble cannot.

ETHICS AND PRIVACY STATEMENT

This study uses only publicly available market data obtained from the public REST endpoints of centralized cryptocurrency exchanges. The collected records are aggregate market observations—quotes, aggregated order-book depth, anonymous trade prints, funding rates, and open interest—and contain no personal data, no account identifiers, and no information attributable to any individual trader. No human subjects were involved and no institutional review was required.

Data collection respected each venue’s published rate limits; the sequential polling design that produces our approximately 110-second cadence is a direct consequence of staying within those limits rather than circumventing them. No authenticated, paid, or otherwise access-restricted endpoint was used, and no exchange terms of service were bypassed.

All trading in this work is simulated. No orders were submitted to any venue, no capital was deployed, and the reported profit figures are proxy quantities in z-score units rather than realized returns. We emphasize this because research on trading strategies can be misread as investment advice: as documented in Section 7.3, the strategies evaluated here are *not* profitable once realistic transaction costs are applied, and nothing in this paper should be construed as a recommendation to trade.

We also acknowledge the broader externalities of research into market inefficiencies. Cross-exchange latency arbitrage transfers value from slower venues and their participants to faster ones, and

publishing such findings can accelerate that transfer. We judge the disclosure to be net beneficial because the inefficiencies we document are structural properties of a fragmented market that are already visible to well-resourced participants, and because transparency about which signals do *not* survive costs is more likely to deter uninformed retail imitation than to enable it. The accompanying dataset is released to support reproducibility and benchmarking; it contains only the public market observations described above.

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